The Red/Black Game:
Rules, Deterministic Dynamics, and Exact State Enumeration
(Heads-Up)

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Abstract

Red/Black is a hidden-information bidding game played over a shared 52-card deck in which only color matters. This paper gives a complete formal account of the two-player game as a *deterministic, acyclic finite transition system*: we specify the state set, input alphabet, transition function, initial states and final states, and we count each of them exactly. Three results organize the account. First, once the deal is fixed the game is fully deterministic — all chance is confined to the initial state — and the transition graph is acyclic, so every play terminates in between 5 and 9 rounds. Second, the order in which cards are flipped during a contest is irrelevant to the state, which collapses procurement from a tree of flip sequences to a small lattice on two counters. Third, the public projection of the state — what an observer knows — reduces to two integers per seat, giving 6,051 public classes, of which exactly 5,485 are reachable; the pruning is concentrated almost entirely in one layer, where 59% of the enumerable classes cannot occur. We then characterize what that projection throws away: it pins down the support of the public posterior exactly, but not its shape, so two histories arriving at the same public state can leave an observer with materially different beliefs. We then measure the object the projection summarizes: the real game's public history tree has 5.3×10^{34} decision histories and exactly 2,448,351,139,319,171,077,272,326,424,233,304 perfect-recall infosets — thirty orders of magnitude beyond the class space — so no method can engage it by tabulation. Finally we prove that the beliefs the class forgets are not formless: under any strategy profile, every reachable posterior factorizes over the class prior into two per-seat tilts, giving the game a finite-dimensional belief state whose shape is strategy-independent. Every count in this paper is machine-verified against the shipped rules engine. Two companion papers build on it: one solves the class abstraction exactly and measures what the projection costs in play; the other uses the belief family to engage the real game.

1 Introduction

Red/Black is a bidding game for two to ten players. A shared 52-card deck is dealt face down; only the color of a card matters, and there are exactly 26 red and 26 black. Players bid on how many cards of a chosen color are in play across *all* hands. An opponent either raises the bid or challenges it, and a challenged bidder must literally produce the claimed cards by flipping them, one at a time, off the top of any player's deck. Fail, and you discard a card. Run out of cards and you are eliminated.

The game is small enough to be understood completely and large enough to be interesting. This paper does the first half of that job: it describes the game exactly, formalizes it as a finite transition system, and counts everything. It deliberately says nothing about how to *play* well. Strategy, equilibrium and the algorithms that approximate it are the subject of two companion papers — Paper II solves the class abstraction exactly and measures its cost; Paper III engages the real game — and here we build the object those algorithms run on.

Scope. We restrict throughout to the **heads-up** ($N = 2$) game. The rules in Section 2 are stated for two players; where a rule generalizes awkwardly we say so. All counting is heads-up.

Contributions.

1. A complete formal specification of Red/Black as a deterministic transition system, with an explicit account of where the randomness lives and what the automaton does *not* capture (Section 3).
2. Termination and acyclicity theorems: the heads-up game lasts between 5 and 9 rounds, and no state is ever revisited (Theorems 3.7 and 3.8).
3. An order-irrelevance lemma for procurement (Lemma 4.3) that collapses the contest phase from a tree of flip sequences to a lattice on two counters, and the exact node counts that follow (Section 4).
4. Exact enumeration of the public state space: a closed form $\binom{H+3}{3} - 1$ for the per-seat public statistic, hence 6,051 public classes at hand size $H = 5$ (Section 8).
5. An exact reachability closure showing only 5,485 of those 6,051 classes can occur, with the obstruction identified and proved (Section 9).
6. A characterization of what the public state *omits*: the bounds describe the support of the public posterior exactly (Theorem 11.1) but not its shape, and two histories reaching the same class can leave an observer with materially different beliefs (Section 11).
7. Exact enumeration of the *real* game: 5.3×10^{34} public decision histories and 2.4×10^{33} perfect-recall infosets at $H = 5$, counted exactly (not estimated) by a recurrence whose opponent-realizability test is Theorem 11.1 itself, validated by explicit joint enumeration at small hand sizes (Section 10).
8. The rank-one factorization of reachable beliefs: under any strategy profile the public posterior at a boundary is the class prior times two per-seat tilts (Theorem 11.3), with opponent-model-free own-side conditionals (Corollary 11.4) — the structural fact the companion papers' belief tracking and safe re-solving stand on (Section 11.4).

Reproducibility. Every numeral in this paper is produced by `state_counts.py`, which rederives the combinatorics from the rules independently of the game engine, and cross-checked against the shipped engine by `verify_against_engine.py`. Both accompany this document.

Road map. Section 2 states the rules. Section 3 builds the automaton. We then climb a ladder of three games, identical in rules and differing only in how many cards each player is dealt: the one-card game (Section 6), small enough to draw in full; the two-card game (Section 7), the smallest game with a real bidding ladder and a real discard choice; and the actual five-card game (Section 8). Section 9 settles reachability, Section 11 examines what the public state deliberately throws away, and Section 13 hands off to the companion paper.

Notation. Symbols are fixed for the whole paper and collected in Table 1. Seats are always 0 and 1, and i always ranges over them. Two conventions prevent collisions: lower bounds always carry the lb subscript, so b_i (blacks held) is never confused with b^i_{lb} (blacks known to be held);

and bid counts are written k or κ, never b. A primed symbol always means "after the pending discard" — the distinction is the subject of Section 5.2, and it is the one place where the rules' ordering and the analysis' ordering disagree.

Table 1: Notation used throughout.

Symbol	Meaning	Defined
N	number of players; $N = 2$ throughout	§2
H	hand size at the deal, and the most a seat can ever hold	§2
n_i	cards in seat i's hand *now* (at a boundary: pre-discard)	Def. 3.1
T	cards in play now, $T = n_0 + n_1$	Def. 3.1
n'_i	seat i's hand size *during* the next round (post-discard)	§5.2
T'	cards in play during the next round; the **layer index**	§5.2
$r_i,\ b_i$	reds and blacks currently in seat i's hand, $r_i + b_i = n_i$	§9.2
ρ	phase	Def. 3.1
σ	seat that starts the current round	Def. 3.1
h_i	seat i's hand as an *ordered* tuple of colors	Def. 3.1
β	bid history: a color plus a chain $k_1 < \cdots < k_m$	Def. 3.1
$\nu,\ \lambda$	the bidder; the loser of the contest just resolved	Def. 3.1
φ_i	cards flipped off seat i's deck this contest	Def. 3.1
c	the bid color of the current round	§2
$k,\ \kappa$	a bid count; the *final* count of a chain ($\kappa = k_m$)	§2
$x,\ y$	flips taken off the bidder's / the challenger's deck	Lem. 4.6
L	lattice-point count for procurement	Lem. 4.6
ℓ	seat owing a discard at a boundary; -1 at the root	Def. 5.1
s_i	seat i's public statistic $(n_i, r^i_{\mathrm{lb}}, b^i_{\mathrm{lb}})$	Def. 5.1
$r^i_{\mathrm{lb}},\ b^i_{\mathrm{lb}}$	certainty bounds: reds / blacks seat i surely holds	Def. 5.1
$K(H)$	number of distinct per-seat public statistics	Thm. 5.6
$m_i,\ e_i$	on-color / off-color cards revealed from seat i in a round	Lem. 9.2

2 The Rules

2.1 Components and setup

One shared deck of 52 cards, exactly 26 red and 26 black. Suits and ranks do not exist: a card's only attribute is its color. Each of the two players is dealt $H = 5$ cards; the remaining $52 - 2H = 42$ cards are removed from the game and never seen by anyone. Hands only ever shrink — no card is drawn after the deal.

A player's hand is an *ordered* list. Index 0 is the top of the deck: the first card that will be flipped if someone procures from it. Players see their own cards and nothing of their opponent's except its size, which is always public.

2.2 A round

A round has four phases, in order.

1. **Reorder.** Both players privately and simultaneously rearrange their own hand and confirm. This is the only moment hand order can change.

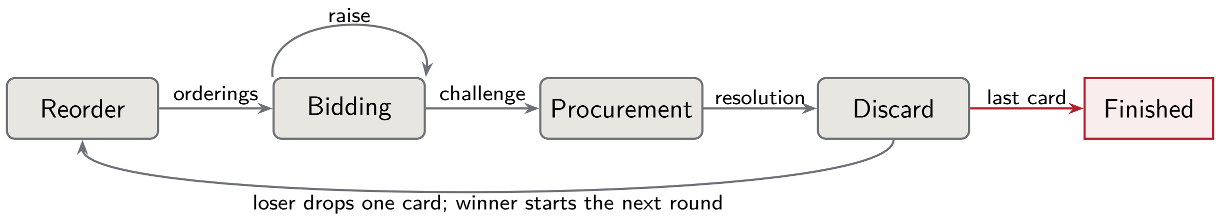

Figure 1: The four phases of a round. The only cycle in the *phase* graph is the round loop; it is not a cycle in the state graph, because one card leaves play on every pass (Theorem 3.7).

2. **Bidding.** The starting player bids a pair (k, c): a count $k \geq 1$ and a color c, claiming that at least k cards of color c are in play across both hands. The count is capped by the total number of cards in play. Thereafter players alternate, and each must do exactly one of: *raise* (same color, strictly greater count) or *challenge*. The color is fixed by the first bid and never changes within a round. Passing is not a legal move.

3. **Procurement.** A challenge ends bidding. The last bidder must now produce the claim. They repeatedly flip the top card of *either* player's deck — their own or the opponent's, switching freely — and each flipped color is revealed publicly. The first off-color flip is fatal: the bidder loses immediately. Reaching the claimed count of on-color cards wins: the challenger loses.

4. **Discard.** The loser of the contest secretly removes one card from their hand. The identity of that card is never revealed. If the loser holds exactly one card, the discard is forced and eliminates them.

The winner of the contest starts the next round. In round 1 the starting player is seat 0. A player with zero cards is eliminated and the other player wins.

2.3 Information

What is public at all times: both hand sizes, whose turn it is, the current round's full bid history, and the full flip history of the current contest. What is private: the colors and order of your own hand, and the identity of any card you yourself discarded.

Two rules do most of the work in this paper.

Discard anonymity. The discarded card's identity is never revealed. Observers learn only that a hand shrank by one.

Reorder scrambling. Cards flipped in round k are identifiable only within round k. Once the next reorder happens, all of a player's cards are anonymous card backs again.

Together these mean no *card identity* survives a round boundary. What survives is strictly weaker: multiset knowledge, degraded by discards. If you saw two reds flipped from my hand, you know I hold at least two reds — until I discard, at which point you know only that I hold at least one, because the card I dropped may have been one of them. We make this precise in Section 5.

3 The Game as a Deterministic Transition System

3.1 Is it a finite state machine?

Yes, with one clarification worth making carefully, because the phrase "hidden information" suggests otherwise.

A deterministic finite automaton is a tuple $(\mathcal{Q}, \Sigma, \delta, q_0, F)$ with $\delta : \mathcal{Q} \times \Sigma \to \mathcal{Q}$ a *function* — one input, one successor, no coin flips. Red/Black satisfies this once we are careful about two things:

1. **The randomness is in the initial state, not the transitions.** The deal is the only chance event in the entire game. Fix it, and every subsequent transition is a deterministic function of the current state and the acting player's move. So Red/Black is a *family* of deterministic automata indexed by the deal, or equivalently one automaton whose first input letter is the deal.
2. **Hidden information is a property of the observers, not of the machine.** The state is perfectly well defined at all times; it is simply that neither player can see all of it. Formally, hidden information is an *equivalence relation on* $\mathcal{Q}$ (Section 3.6), layered on top of a machine that is itself deterministic.

So the intuition is right. The machine is deterministic. What a finite state machine does not by itself express is who can distinguish which states — and that is exactly the structure the companion paper needs.

3.2 The state

Definition 3.1 (Game state)**.** A state is a tuple

$$q = \big(\rho,\ \sigma,\ h_0,\ h_1,\ \beta,\ \nu,\ \varphi_0,\ \varphi_1,\ \lambda\big)$$

with components

ρ	phase, $\rho \in \{\textsc{reorder}, \textsc{bid}, \textsc{proc}, \textsc{disc}, \textsc{fin}\}$
σ	starting seat of the current round, $\sigma \in \{0, 1\}$
h_i	seat i's hand: an ordered tuple of colors, $h_i \in \{\mathsf{R}, \mathsf{B}\}^{n_i}$
β	bid history: $\perp$, or a pair $(c,\ k_1 < \cdots < k_m)$ of a color and a chain of counts
ν	the bidder (last raiser) once a challenge has occurred, else $\perp$
φ_i	cards already flipped from seat i's deck this contest
λ	the loser of the contest just resolved, else $\perp$

We write $n_i = |h_i|$ for hand sizes and $T = n_0 + n_1$ for the total cards in play.

Remark 3.2 (What is deliberately absent)*.* The state carries no record of *which* cards have been discarded, nor of past rounds' bids or flips. Neither affects any player's legal moves or any future transition: discarded cards are gone from play, and a past round's bid history is cleared. They affect only what the players *believe*, which is the subject of Section 3.6 and of the companion paper. Keeping them out of $\mathcal{Q}$ is what makes the state space finite and small.

3.3 The input alphabet

Definition 3.3 (Inputs)**.** $\Sigma = \Sigma_{\text{deal}} \cup \Sigma_{\text{ord}} \cup \Sigma_{\text{bid}} \cup \Sigma_{\text{proc}} \cup \Sigma_{\text{disc}}$, where

Phase	Letters available	How many
—	$\texttt{deal}(D)$, a split of $2H$ cards	$\binom{52}{H}\binom{52-H}{H}$
REORDER	$\texttt{order}(\pi)$, an ordering of one's own hand	$\binom{n_i}{r_i}$ per seat
BID	$\texttt{bid}(c, k)$, $c \in \{\mathsf{R}, \mathsf{B}\}$, $1 \le k \le T$	$2T$
BID	$\texttt{raise}(k)$, k above the current count	$\le T - 1$
BID	`challenge`	1
PROC	$\texttt{flip}(j)$, j the target seat	≤ 2
DISC	$\texttt{drop}(c)$, $c \in \{\mathsf{R}, \mathsf{B}\}$	≤ 2

Opening bids and raises are mutually exclusive: `bid` is legal only on an empty history, `raise` and `challenge` only on a non-empty one.

δ is a *partial* function: illegal letters have no successor. One may complete it in the usual way with an absorbing dead state; we leave it partial, since the legal-move set is itself part of the game's description.

3.4 Initial and final states

Definition 3.4 (Initial states)**.** q_0 is the pre-deal state. After the single chance letter $\texttt{deal}(D)$ the machine occupies $\big(\text{REORDER}, 0, h_0, h_1, \bot, \bot, 0, 0, \bot\big)$ with $|h_0| = |h_1| = H$. Seat 0 starts round 1 by convention.

Definition 3.5 (Final states)**.** F is the set of states with $\rho = \text{FIN}$, equivalently those in which some $n_i = 0$. The other player is the winner. There are no draws: exactly one player is eliminated, and the game ends the moment that happens.

3.5 Determinism, acyclicity, termination

Theorem 3.6 (Determinism)**.** *For every state q and every legal letter a, the successor $\delta(q, a)$ is uniquely determined. No transition consults a random source.*

Proof. Inspect each phase. REORDER replaces h_i by a stated permutation. BID appends to β or moves to PROC and records ν. In PROC, the letter names a target seat j; the card revealed is $h_j[\varphi_j]$, already fixed in the state, and the outcome is a comparison of that color against β's color: off-color $\Rightarrow$ bidder loses, else increment φ_j and check the count. DISC names a color and removes one such card. In every case the successor is a function of (q, a) alone. □

The subtle point is in PROC: the flipped card looks random to the players, but it is not random to the machine. It was fixed at the deal and positioned at the last reorder. Uncertainty here is epistemic, not stochastic.

Theorem 3.7 (Acyclicity)**.** *The transition graph is acyclic.*

Proof. Take the lexicographic rank $\Phi(q) = \big(T,\ -|\beta|,\ -(\varphi_0 + \varphi_1)\big)$ ordered so that every transition strictly decreases it. Within a contest, each `flip` increments $\varphi_0 + \varphi_1$ and leaves T and $|\beta|$ fixed. Within bidding, each `bid` or `raise` increments $|\beta|$ and leaves T fixed; `challenge` moves to PROC with $\varphi_0 + \varphi_1 = 0$, which is handled by treating phase order as a tiebreak. Each `drop` decrements T by exactly one, resetting the later components. As T strictly decreases once per round and never increases, no state recurs. □

Theorem 3.8 (Round bounds)**.** *The heads-up game with hand size H lasts at least H and at most $2H - 1$ rounds. For the real game ($H = 5$) that is between 5 and 9 rounds.*

Proof. Exactly one card leaves play per round, and a seat is eliminated on its H-th discard. The minimum is achieved when the same seat loses every round: H rounds. For the maximum, the game is still alive after k rounds only if both seats have made at most $H - 1$ discards, so $k \leq 2(H-1)$; the game therefore ends on round $2H-1$ at the latest, and that bound is achieved by alternating losses. □

Corollary 3.9. *Total cards in play is a strictly decreasing round counter running $2H$, $2H-1$, … down to the terminal value, so it is a natural index for induction over the game. Section 5.2 fixes precisely which version of that count we index by — the one measured during a round rather than at its boundary — and calls it T'.*

3.6 What the automaton omits: the information partition

Definition 3.10 (Observation and indistinguishability)**.** Let $\mathrm{obs}_i(q)$ be the part of q visible to seat i: the phase, the starter, both hand sizes, the current bid history β, the flip counters and the colors revealed so far this contest, plus seat i's own hand h_i. Write $q \sim_i q'$ when $\mathrm{obs}_i(q) = \mathrm{obs}_i(q')$. The classes of $\sim_i$ are seat i's *information sets.*

$\sim_i$ is what makes Red/Black a game rather than a puzzle, and it is invisible to $(\mathcal{Q}, \Sigma, \delta, q_0, F)$. Two facts about it matter here:

1. **Perfect recall** holds: a player never forgets their own past observations or moves. This is what licenses the algorithms of the companion paper.
2. **The public projection**, $\mathrm{pub}(q)$ — everything both players see — is far smaller than q, and at round boundaries it collapses to just four integers. Section 5 derives it, and it is the object the whole second half of this paper counts.

4 The Round Automaton

Before climbing the ladder of games we isolate the structure of a *single* round, which is independent of the deal and depends only on the starter σ and the hand sizes (n_0, n_1).

4.1 Bidding

A bid history is a color together with a strictly increasing chain of counts drawn from $\{1, \ldots, T\}$. Chains are in bijection with non-empty subsets of $\{1, \ldots, T\}$.

Lemma 4.1 (Bid node count)**.** *The number of bidding states with T cards in play is*

$$1 \; + \; 2\left(2^T - 1\right) \; = \; 2^{T+1} - 1.$$

Proof. One pre-bid state; then a state per (color, non-empty chain) pair, and there are $2^T - 1$ non-empty subsets of $\{1, \ldots, T\}$ and two colors. □

So $T = 2$ gives 7 states, $T = 4$ gives 31, and the full game's opening round ($T = 10$) gives 2,047. The seat to act at a chain of length m is σ if m is even and $1 - \sigma$ otherwise; the bidder — the seat that will have to produce the claim — is whoever placed the last bid.

Remark 4.2 (A constraint we will need). Because the starter bids first and counts strictly increase, a *non-starter* can only be the bidder if at least two bids were placed, which forces the final count to be at least 2. A final count of 1 pins the bidder to be the starter. This small fact prunes real states in Section 9.

4.2 Procurement, and why order does not matter

Lemma 4.3 (Order irrelevance)**.** *During a contest, the state depends only on the pair of counters* (φ_0, φ_1) *and not on the order in which the flips were made.*

Proof. A contest continues only while every flipped card has matched the bid color. Hence at any live procurement state, the multiset of revealed cards is determined: $\varphi_0 + \varphi_1$ cards, all of the bid color, of which φ_i came off seat i's deck. Each deck is consumed strictly top down, so which cards remain is likewise determined by the counters alone. Two flip sequences with the same counters therefore yield identical states. □

Lemma 4.3 is what makes the game tractable: procurement is a *lattice* on two counters, not a tree of flip sequences. A naive sequence count at $T = 10$ would be exponential in the number of flips; the lattice is quadratic.

A second, independent collapse applies to the *orderings* themselves.

Lemma 4.4 (Hand orderings collapse at the first color change)**.** *Let a seat's post-reorder hand open with a maximal run of color a of length k. Then nothing at position $k+2$ or beyond can ever be revealed, under either bid color. Consequently two orderings are strategically indistinguishable whenever they agree on the pair*

$$(a,\ k) \ = \ (\textit{leading color},\ \textit{leading run length}),$$

and the number of distinguishable orderings of an $(r,\ b)$ multiset is

$$F(r, b) \ = \ \begin{cases} 1, & r = 0 \ \textit{or} \ b = 0, \\ r + b, & \textit{otherwise,} \end{cases}$$

against $\binom{r+b}{r}$ raw orderings.

Proof. Procurement from a seat's deck consumes cards top down and stops at that seat's first off-color card. If the bid color is a, the flips take the k cards of the leading run and then the card at $k + 1$, which is $\neg a$ by maximality, and which is fatal. If the bid color is $\neg a$, the very first flip is off-color and fatal. In both cases consumption halts at position $k + 1$ at the latest, so positions $k + 2, \ldots, n$ are never seen.

Nor can they matter later. Flipping does not remove a card (Section 2), so a round leaves hand *composition* untouched; and every round begins with a fresh reorder, so no ordering survives a boundary. Hence the tail is unobservable and irrelevant, and the observable content of an ordering is exactly (a, k).

For the count: with both colors present, the leading run may be $a =$ red of length $1..r$ or $a =$ black of length $1..b$, giving $r + b$ classes; a single-color hand admits one. □

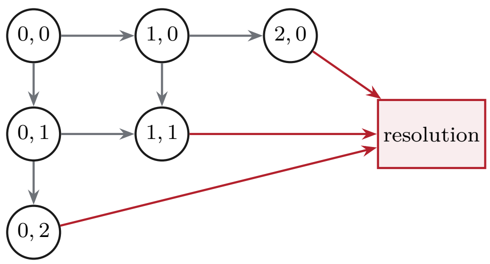

Figure 2: The procurement lattice for a final bid count of $\kappa = 3$ with two cards in each hand. A state is the pair (x, y) counting cards already flipped off the bidder's and the challenger's decks; by Lemma 4.3 the path taken to a node is irrelevant. Only the $x+y \le \kappa - 1 = 2$ nodes are live.

Remark 4.5 (Size of the collapse). The saving needs $n \ge 4$ to appear at all — for $n \le 3$, $\binom{n}{r} = F$ exactly — and is capped at 50% per composition ($\binom{5}{2} = 10 \to 5$). Balanced hands save; lopsided ones ($r \in \{0, 1, n-1, n\}$) save nothing. Summed over every reachable private state of the full game the reduction is 12% (172,488 $\to$ 150,964), but it is worth more than that in practice because it grows with depth — 0% at $T' \le 4$, 15% at $T' = 8$, 24% at $T' = 9$, 31% at the root — and the deep layers hold most of the work. Weighted by how many decisions actually sit in each layer it is about 19%; the companion paper tabulates those weights.

Note this is a *lossless* collapse, like Lemma 4.3 and unlike the belief abstraction of Section 11: merged orderings are indistinguishable in every continuation, so nothing is approximated.

Lemma 4.6 (Procurement node count)**.** *Fix a color and a chain $k_1 < \cdots < k_m$, and write n_{bid} and n_{chal} for the hand sizes of the bidder and the challenger. The live procurement states for that chain number*

$$L\big(k_m - 1,\ n_{\text{bid}},\ n_{\text{chal}}\big) \ = \ \#\big\{(x, y) \in \mathbb{Z}^2_{\ge 0} \ : \ x + y \le k_m - 1,\ x \le n_{\text{bid}},\ y \le n_{\text{chal}}\big\},$$

where x and y count the cards already flipped off the bidder's and the challenger's decks respectively. Summing over both colors and all $2^{k_m - 1}$ chains with final count k_m gives the round's total. When $n_0 = n_1 = n$ this is

$$\#\textsc{proc}(T, n) \ = \ 2 \sum_{\kappa=1}^{T} 2^{\kappa - 1} L(\kappa - 1, n, n),$$

the sum running over the possible final counts κ.

The constraint $x + y \le k_m - 1$ is the statement that the bid is not yet made: at $x + y = k_m$ the bidder has produced the claim and the round is over.

4.3 Resolutions

Every contest ends in one of two ways, and the resolution records enough to continue the game: the color and chain, the final counters, whether the loser was the bidder, and — when the bidder busted — which deck the fatal card came off. That last datum matters because it tells observers whose hand certainly contains an off-color card.

Table 2 collects the three counts for the hand-size configurations used later.

Table 2: The round automaton, per (n_0, n_1). Counts are independent of the starter, and every entry is verified against the shipped graph builder.

Configuration		T	Bid states	Procurement states	Resolutions
$(1, 1)$	one-card game	2	7	14	28
$(2, 2)$	two-card game	4	31	190	348
$(5, 5)$	opening round	10	2,047	65,790	119,036
$(3, 5)$	a later round	8	511	10,398	18,588

5 The Public State at a Round Boundary

We now derive the projection $\mathrm{pub}(q)$ that both players share. It is the single most useful object in the paper.

5.1 What survives a round

Fix the moment a contest resolves: the loser is known, but has not yet discarded. Call this a *boundary.* By the rules of Section 2.3, an observer at a boundary knows: both hand sizes, who lost, and — from the flips just seen, plus memory of earlier rounds — a lower bound on how many cards of each color each seat certainly still holds. Nothing else about cards survives, because discard anonymity and reorder scrambling destroy identity.

Definition 5.1 (Public class)**.** A boundary's public state is

$$\big(\ell,\ s_0,\ s_1\big), \qquad s_i = \big(n_i,\ r^i_{\mathrm{lb}},\ b^i_{\mathrm{lb}}\big),$$

where $\ell \in \{-1, 0, 1\}$ names the seat that owes a discard (-1 at the game root), n_i is seat i's hand size *before* that discard, and $r^i_{\mathrm{lb}}, b^i_{\mathrm{lb}}$ are the certainty bounds: everyone knows seat i holds at least r^i_{lb} reds and at least b^i_{lb} blacks. The next round's starter is $1 - \ell$, so it need not be stored.

Lemma 5.2 (Bound folding)**.** *Across a round, the bounds evolve as*

$$r_{\mathrm{lb}} \longmapsto \max\big(r_{\mathrm{lb}} - [\,\textit{this seat discarded}],\ \#\{\textit{reds seen from this seat this round}\}\,\big),$$

and symmetrically for b_{lb}*. The fold is a maximum, never a sum.*

Proof. A discard removes an unknown card, so a bound of r_{lb} degrades to $r_{\mathrm{lb}} - 1$ (floored at zero): the dropped card may have been one of the counted reds. Cards flipped this round are certainly still held, giving a fresh bound. The two cannot be added, because a red seen this round may be one of the reds already counted; only the larger is certain. $\square$

Lemma 5.3 (Certainty is bounded by holdings)**.** $r^i_{\mathrm{lb}} + b^i_{\mathrm{lb}} \le n_i$ *at every boundary.*

5.2 Two clocks: where the discard sits

There is one genuine offset in this paper, and it causes trouble if left implicit, so we make it explicit here and use it consistently afterwards.

The rules order a round as reorder → bidding → procurement → *discard*: the discard is the *last* thing that happens (Section 2, Figure 1). That is the right description of the game as played.

But the natural place to cut the game is not the same place. We cut at a *boundary* — a procurement resolution, where the loser is known but has not yet discarded — because that is the moment at which the public state is a clean summary. Cutting there means the segment between two consecutive boundaries reads *discard* → reorder → bidding → procurement: the discard is now the *first* thing that happens.

Nothing about the game changed. The same events occur in the same order; only the cut points moved, by exactly one discard.

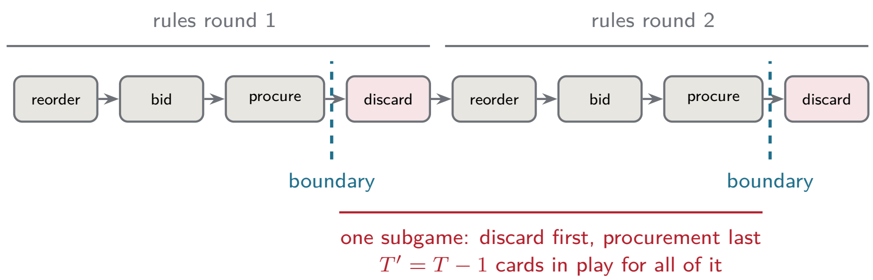

Figure 3: The same event tape, cut two ways. The rules bracket a round from reorder to discard (grey, above). The analysis brackets a subgame from one procurement resolution to the next (red, below), so the discard moves to the front. A boundary sits between procurement and discard, which is why a boundary's hand sizes n_i are *pre*-discard while the round that follows is played with n'_i.

Definition 5.4 (In-round sizes and the layer index)**.** At a boundary with pending discarder ℓ, write

$$n'_i \;=\; n_i - [\, i = \ell \,], \qquad T' \;=\; n'_0 + n'_1 \;=\; T - [\, \ell \geq 0 \,].$$

So n_i and T describe the table at the boundary; n'_i and T' describe it for the entire round that follows, since no card leaves play between the discard and the next resolution. We index classes by T', the **layer**.

Remark 5.5 (Why the layer is T' and not T)*.* Two boundaries with the same T can have different amounts of play left: the game root has $T = 10$ with no discard pending, whereas a resolution at $n_0 = n_1 = 5$ also has $T = 10$ but owes a discard, so only 9 cards will actually be bid over. Indexing by T' puts them in different layers, which is what makes the class space a clean DAG with exactly one edge per round (Figure 7). Indexing by T would not.

Two consequences worth stating plainly, because both are easy to misread:

- A boundary's hand sizes n_i are *pre*-discard. A seat listed with $n_i = 1$ is about to be eliminated, not already gone.
- Everything in Section 4, the round automaton, is expressed in the sizes that hold *during* the round — so a round reached from a boundary uses n'_i and T' there, not n_i and T. Read in isolation, Section 4 describes a generic round and the primes are unnecessary; read as the successor of a boundary, they matter.

5.3 Counting public classes

Theorem 5.6 (Per-seat statistic)**.** *For hand size H, the number of distinct values of $(n, r_{\text{lb}}, b_{\text{lb}})$ with $1 \le n \le H$ and $r_{\text{lb}} + b_{\text{lb}} \le n$ is*

$$K(H) \;=\; \sum_{n=1}^{H} \binom{n+2}{2} \;=\; \binom{H+3}{3} - 1.$$

Proof. For fixed n, the pairs $(r_{\text{lb}}, b_{\text{lb}})$ with $r_{\text{lb}} + b_{\text{lb}} \le n$ number $\binom{n+2}{2}$ (weak compositions of at most n into two parts). The hockey-stick identity gives $\sum_{n=0}^{H} \binom{n+2}{2} = \binom{H+3}{3}$; dropping the $n = 0$ term, worth 1, gives the claim. □

So $K(1) = 3$, $K(2) = 9$, $K(5) = 55$.

Corollary 5.7 (Class count)**.** *The number of public classes is $2K(H)^2 + 1$: two choices of the seat owing a discard, an independent statistic per seat, plus the game root. For $H = 5$ that is $2 \cdot 55^2 + 1 = 6{,}051$.*

Corollary 5.8 (Terminal classes)**.** *A boundary whose pending loser holds one card is terminal: the discard is forced and eliminates them. There are $2 \cdot 3 \cdot K(H)$ such classes, or 330 at $H = 5$.*

Classes are indexed by the layer T' of Definition 5.4. By Theorem 3.7, T' strictly decreases from one boundary to the next, so the class space is a layered directed acyclic graph.

6 Warm-Up I: The One-Card Game

Deal each player a single card from the real 52-card deck. Everything else is unchanged. The game lasts exactly one round (Theorem 3.8 with $H = 1$), and it is small enough to write down completely.

6.1 Deals

There are $52 \cdot 51 = 2{,}652$ card-level deals. Only color matters, so strategically there are 4 outcomes, with

$$\Pr[\text{both red}] = \Pr[\text{both black}] = \tfrac{26\cdot 25}{52\cdot 51} = \tfrac{25}{102}, \qquad \Pr[\text{different}] = \tfrac{2\cdot 26\cdot 26}{52\cdot 51} = \tfrac{26}{51}.$$

Note $\Pr[\text{different}] = \frac{26}{51} \approx 0.5098 > \frac{1}{2}$: the two cards are slightly *anti*-correlated, because they are drawn without replacement from a balanced deck. This tiny asymmetry is not an artefact; it is the seed of all card-counting in the full game.

6.2 The automaton in full

With $T = 2$, Lemma 4.1 gives $2^3 - 1 = 7$ bidding states. The opener may claim 1 or 2 of either color. A claim of 2 cannot be raised (the count is already at the cap), so it must be challenged. A claim of 1 may be raised to 2, after which the opener must challenge. Figure 4 draws the whole bidding phase for one color; the other color is an exact mirror.

Procurement adds 14 states and 28 distinct resolutions (Table 2). A claim of 1 resolves on the very first flip. A claim of 2 admits the three counter pairs $(0,0), (1,0), (0,1)$ before resolving.

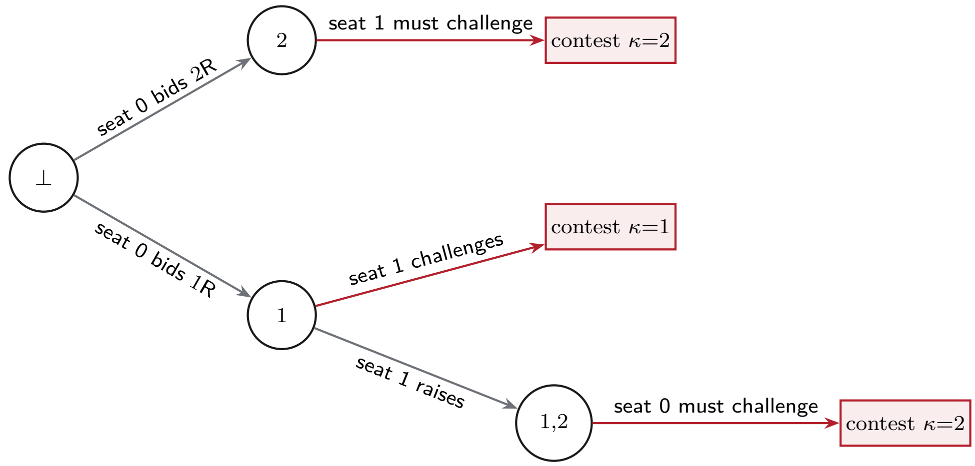

Figure 4: The complete bidding phase of the one-card game, red claims only (black is the mirror). Node labels are the chain of counts; $\perp$ is the pre-bid state. Together with the black mirror this is $1 + 2 \cdot 3 = 7$ states, matching Lemma 4.1.

6.3 Public classes

With $H = 1$, Theorem 5.6 gives $K(1) = 3$ per-seat statistics — $(1, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$ — and Corollary 5.7 gives $2 \cdot 9 + 1 = 19$ classes. Exhaustive forward search (Section 9) finds 28 concrete boundary states realizing 17 of those 19.

Example 6.1 (The two impossible classes)**.** The two unreachable classes are $s_0 = s_1 = (1, 0, 0)$, for either loser: a boundary at which *nothing at all* is publicly known about either card. It cannot occur, because a contest cannot resolve without at least one flip — the bidder must either produce a card or bust on one — and every flip is public. This is Lemma 9.1.

It is worth noting what is *not* excluded. One might guess that $s_0 = s_1 = (1, 0, 1)$ — both seats known to hold a black — is impossible, on the grounds that under a red claim only one black can ever be revealed. But the claim need not be red: under a *black* claim of 2, the bidder flips one black off each deck, makes the bid, and both bounds rise together. The class is reachable. Fixing the bid color too early is the standard way to get these arguments wrong, and we return to it in Remark 9.5.

7 Warm-Up II: The Two-Card Game

Now deal two cards each. Three genuinely new things appear: a bidding ladder with room to maneuver, a discard that is a real *choice* rather than a forced elimination, and therefore a second round — the first time the game recurses into a smaller copy of itself.

7.1 Deals and rounds

There are $\binom{52}{2}\binom{50}{2} = 1{,}624{,}350$ card-level deals and 9 color outcomes $(r_0, r_1) \in \{0, 1, 2\}^2$. By Theorem 3.8 the game lasts 2 or 3 rounds.

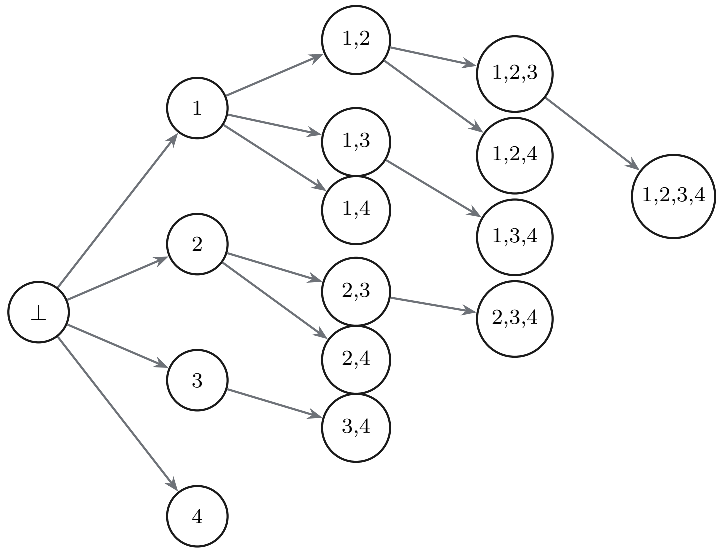

Figure 5: The bid-chain lattice for one color at $T = 4$: chains are non-empty subsets of $\{1, 2, 3, 4\}$, so there are $2^4 - 1 = 15$ per color and $2 \cdot 15 + 1 = 31$ bidding states in total. Every node also has a *challenge* edge into procurement, omitted here.

7.2 Structure

With $T = 4$ the opening round has $2^5 - 1 = 31$ bidding states, 190 procurement states and 348 resolutions. The chain lattice is now non-trivial: an opener claiming 1 can be raised to 2, 3 or 4, each of which can be raised again, and so on.

7.3 The first real discard

At $H = 2$ the loser of round 1 holds two cards and must choose which to drop. This is the game's first genuinely private decision with a future: the choice is unobserved, it determines the loser's hand for round 2, and — by discard anonymity — the opponent will never learn it directly. It is also where the public bounds first decay (Lemma 5.2).

7.4 Class counts

$K(2) = 9$ per-seat statistics give $2 \cdot 81 + 1 = 163$ classes, spread over layers $T = 4$ (the root), $T = 3$, $T = 2$ and $T = 1$. Forward search realizes 149 of them across 345 concrete boundary states; the breakdown is in Table 5.

8 The Full Game

The real game deals $H = 5$. There are

$$\binom{52}{5}\binom{47}{5} = 3{,}986{,}646{,}103{,}440$$

card-level deals and 36 color outcomes $(r_0, r_1) \in \{0, \ldots, 5\}^2$, distributed as in Table 3. The opening round alone has 2,047 bidding states, 65,790 procurement states and 119,036 resolutions.

Table 3: The color-deal law at $H = 5$: $\Pr[r_0 = u,\ r_1 = v]$, where r_i is the number of reds dealt to seat i. Rows are u (seat 0), columns v (seat 1). Mass concentrates on balanced hands, and the slight anti-diagonal tilt is the without-replacement effect.

	$v = 0$	1	2	3	4	5
$u = 0$	0.00034	0.00257	0.00713	0.00901	0.00518	0.00109
1	0.00257	0.01783	0.04505	0.05180	0.02713	0.00518
2	0.00713	0.04505	0.10360	0.10854	0.05180	0.00901
3	0.00901	0.05180	0.10854	0.10360	0.04505	0.00713
4	0.00518	0.02713	0.05180	0.04505	0.01783	0.00257
5	0.00109	0.00518	0.00901	0.00713	0.00257	0.00034

Table 4: Public classes at $H = 5$ by layer T' — the cards in play *during* the round that follows, i.e. after the pending discard (Definition 5.4).

T'	With a subgame	Terminal	Total
1	0	18	18
2	36	36	72
3	132	60	192
4	330	90	420
5	686	126	812
6	1,104	0	1,104
7	1,290	0	1,290
8	1,260	0	1,260
9	882	0	882
10	1 (root)	0	1
Total	5,721	330	6,051

By Corollary 5.7 there are 6,051 public classes. Table 4 breaks them down by layer, separating the 330 terminal classes — where the pending loser holds one card, so the value is decided without further play — from the 5,721 that contain a real subgame.

9 Reachability

Corollary 5.7 counts the classes the *definition* admits. Not all of them can occur. This section characterises the obstruction, proves it, and reports the exact reachable set.

9.1 The obstruction

Lemma 9.1 (Every round reveals something)**.** *At least one card is flipped in every contest, so at every non-root boundary at least one of the four bounds is positive.*

Proof. The claim's count is at least 1. The bidder must keep flipping until either the count is reached — which requires at least one on-color card to be revealed — or an off-color card appears, which is itself a flip. Either way at least one card is revealed, and by Lemma 5.2 the corresponding bound is raised to at least 1. ☐

Lemma 9.2 (Single-round sighting shape)**.** *Within one round, let c be the bid color. Then the cards revealed from seat i's deck consist of $m_i \geq 0$ cards of color c together with $e_i \in \{0, 1\}$ cards of the other color, and moreover $e_0 + e_1 \leq 1$.*

Proof. Exactly one color is bid per round. Procurement stops the instant an off-color card is flipped, so across the whole round at most one off-color card is ever revealed, and it is the final flip. It comes off exactly one seat's deck, which gives $e_0 + e_1 \leq 1$ and forces every other revealed card to be of color c. ☐

Corollary 9.3 (One round of history)**.** *At a boundary reached after exactly one round — layer $T' = 2H - 1$, which is $T' = 9$ in the full game — the bounds satisfy, for some common color c:*

1. $\min(r^i_{\text{lb}}, b^i_{\text{lb}}) \leq 1$ *for each seat;*
2. *both seats' bounds are expressed against the same color c;*
3. *at most one seat has a non-zero bound on the off color.*

Proof. Immediate from Lemma 9.2 and Lemma 5.2: with only one round played there is nothing to fold against, so the bounds *are* the sightings. ☐

Example 9.4 (Three cannot-happen states at $T' = 9$)**.** Corollary 9.3 rules out, among others:

Class	Status	Why
$s_0 = (5, 3, 1)$, $s_1 = (5, 0, 0)$	reachable	red bid, 3 reds then a fatal black, all from seat 0
$s_0 = (5, 2, 2)$, $s_1 = (5, 0, 0)$	impossible	needs two off-color reveals (part 1)
$s_0 = (5, 3, 1)$, $s_1 = (5, 2, 1)$	impossible	needs two off-color reveals (part 3)
$s_0 = (5, 3, 0)$, $s_1 = (5, 0, 3)$	impossible	needs two different bid colors (part 2)

Remark 9.5 (A trap worth flagging). Reachability arguments of this kind are easy to get wrong by fixing the bid color too early. Consider $\ell = 0$ with $s_0 = (5, 1, 0)$, $s_1 = (5, 0, 0)$: seat 0 owes the discard *and* is known to hold a red. Reasoning under a red bid, one concludes that no off-color card was shown, so the bid was made, so exactly one red was produced, so the final count was 1; by Remark 4.2 the bidder was then the starter, seat 0, and the loser must therefore be seat 1 — contradiction, so the class looks unreachable. It is not. Under a *black* bid, seat 0 opens, immediately flips its own top card, and it is red: the contest ends at once, seat 0 is the loser, and the revealed red is exactly the certainty bound. The class is reachable. This is why we compute the closure by machine rather than argue it by hand.

Note that the obstruction weakens with depth. After two or more rounds, the maximum-fold of Lemma 5.2 can combine sightings under *different* bid colors: a red bid in round 1 revealing $(2, 1)$ followed by a black bid in round 2 revealing $(1, 2)$ folds to $(2, 2)$, which Corollary 9.3 forbids after a single round. So the constraint bites hardest at the top of the game and all but vanishes below.

9.2 Exact closure

To decide which classes occur we search the game forward from the deal. But we cannot search over classes alone, and it is worth being precise about why.

A class records only what is *certain*: that seat i holds at least r_{lb}^i reds. That is not enough to determine which classes can come next, because what a seat can *reveal* in the coming round depends on what it actually holds. A seat whose class says "at least one red" might hold one red or five; only in the latter case can five reds be flipped off its deck next round, driving r_{lb} to 5. So the search must carry the true hand composition alongside the public bounds.

Definition 9.6 (Concrete boundary state)**.** A *concrete boundary state* is

$$\big(\ell,\ (r_0, b_0, r_{\mathrm{lb}}^0, b_{\mathrm{lb}}^0),\ (r_1, b_1, r_{\mathrm{lb}}^1, b_{\mathrm{lb}}^1)\big),$$

where r_i and b_i are the reds and blacks seat i genuinely holds and the remaining entries are as in Definition 5.1. It always satisfies $r_{\mathrm{lb}}^i \le r_i$ and $b_{\mathrm{lb}}^i \le b_i$: you cannot be certain of more than is there.

Each concrete state determines exactly one class — forget r_i and b_i, keep $n_i = r_i + b_i$ and the bounds — so the map

$$\{\text{concrete boundary states}\} \longrightarrow \{\text{classes}\}$$

is a many-to-one projection, and a class is *reachable* precisely when at least one concrete state above it is. This is why the two counts in Table 5 differ so much: 27,990 concrete states at $H = 5$ collapse onto 5,485 classes, roughly five hands per public description.

Example 9.7. The class $\ell = 0$, $s_0 = (4, 1, 0)$, $s_1 = (5, 0, 0)$ says seat 0 owes a discard from four cards and certainly holds a red. Sitting above it are the concrete states with $(r_0, b_0) \in \{(1, 3), (2, 2), (3, 1), (4, 0)\}$ — four different hands, one public description. Distinguishing them is exactly what an opponent cannot do.

The search is then a breadth-first closure: start from every color-deal (r_0, r_1), and from each concrete state generate every state reachable by one legal round — every discard the loser could make, every bid color, every legal final count, either seat as bidder, and every way the contest could resolve, subject to Lemmas 9.1 and 9.2. It terminates because T strictly decreases (Theorem 3.7) and the state set is finite.

One thing is deliberately *not* carried: how many of a seat's own discards were red. By Remark 3.2 that changes no legal move and no public bound, so it cannot affect which classes occur. It does affect what players believe, which is the subject of the next section.

9.3 Reading the result

Two things stand out. First, the reachable set is 90.6% of the enumerable one, so the definition of a public class is tight: the statistics in Definition 5.1 are very nearly a complete invariant. Second, essentially all of the slack sits in a single layer. $T' = 9$ is the only layer with exactly one round of history, and Corollary 9.3 is strongest there; by $T' = 8$ two rounds of max-folding have already restored almost everything.

Table 5: Exact reachability. "Enumerable" counts the classes Definition 5.1 admits; "reachable" counts those realized by some deal and some legal play. All three games are closed exhaustively.

Game	Boundary states	Reachable classes	Enumerable	Pruned
$H = 1$	28	17	19	10.5%
$H = 2$	345	149	163	8.6%
$H = 5$	27,990	5,485	6,051	9.4%

$H = 5$, layer T'	Reachable	Enumerable	Pruned
1	16	18	11.1%
2	68	72	5.6%
3	186	192	3.1%
4	412	420	1.9%
5	802	812	1.2%
6	1,096	1,104	0.7%
7	1,284	1,290	0.5%
8	1,256	1,260	0.3%
9	364	882	**58.7**%
10	1	1	0%
Total	5,485	6,051	9.4%

10 The Size of the Real Game

Everything counted so far is a *state*: a description of where the machine is. But Red/Black is a game of hidden information, and the players do not act on states — they act on what they have *seen*. A bid changes no card and moves no counter, yet it is observed, remembered, and (as Section 11 makes precise) it changes what a rational opponent believes. The decision points of the real game G are therefore *histories*, not states, and this section counts them exactly. The numbers justify, quantitatively, why every solution method in the companion papers works with beliefs rather than with the history tree itself.

10.1 Public histories

Definition 10.1 (Public history)**.** The *public history* at a point of play is the full sequence of publicly observable events since the deal: for each round, a reorder marker, the bid chain as bid (color and count) and challenge events in order, each flip as a (deck, color) pair *in order*, and a discard event naming the discarding seat but — by rule — not the discarded color.

Two features distinguish this object from the states of Section 3. First, bids enter it: the physical state forgets a retracted bid chain the moment the round resolves, but the history keeps every bid ever made. Second, flip *order* enters it: Lemma 4.3 proved order irrelevant to the state, but the sequence of deck choices is observed, and in G an observation is part of the decision point whether or not it moves a counter. Order irrelevance collapses the machine; it does not collapse the information.

The history tree composes round records over the layered DAG of Figure 7, with the loser-dependent starter rule coupling consecutive rounds. Counting it is a finite recurrence over post-

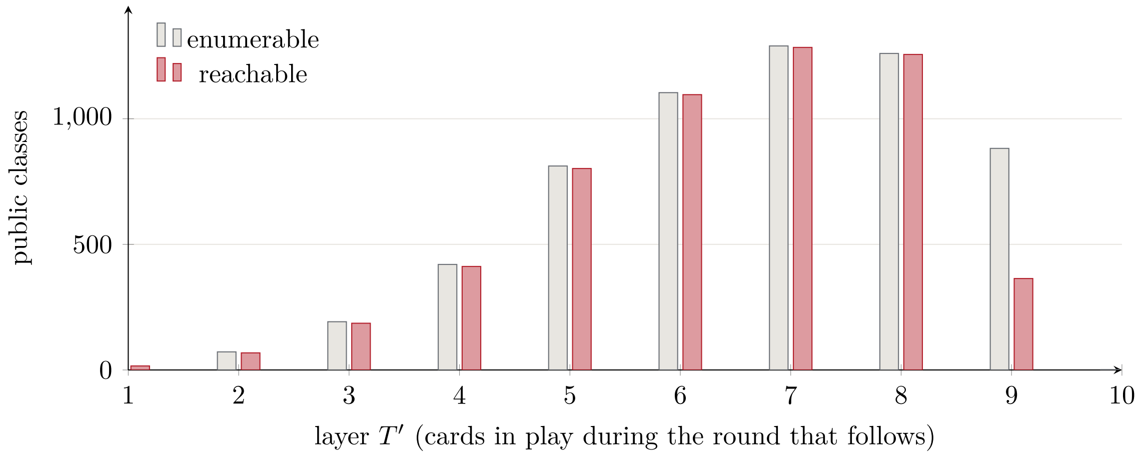

Figure 6: Enumerable versus reachable public classes by layer. The two series are nearly identical everywhere except $T' = 9$ — the layer reached after exactly one round, where Corollary 9.3 eliminates 58.7% of the classes the definition admits.

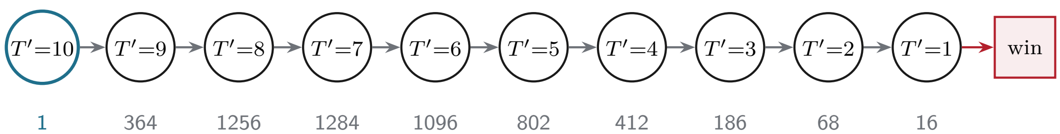

Figure 7: The layered class DAG of the full game. Play enters at the unique root and descends exactly one layer per round, since every round removes exactly one card; the game therefore terminates after between 5 and 9 rounds (Theorem 3.8). Figures beneath each node are the reachable class counts of Table 5.

discard hand sizes (n_0, n_1) and the starter — terminal histories, decision histories (nodes at which some seat acts: a reorder, a bid-or-challenge point, a flip-target point, a discard), and boundary histories are each counted by the same traversal. The recurrence and its closed subterms (seat-capped flip sequences, chain-prefix counts by parity) are implemented in `real_game_size.py`; Table 6 reports the exact results.

At the full hand size the exact values deserve to be written out once, because "10^{34}" undersells the point that they are *counted*, not estimated:

$$\begin{aligned}\text{public decision histories of } G: &\quad 53{,}366{,}481{,}858{,}618{,}743{,}315{,}771{,}794{,}534{,}337{,}111\\ \text{perfect-recall infosets (both seats)}: &\quad 2{,}448{,}351{,}139{,}319{,}171{,}077{,}272{,}326{,}424{,}233{,}304\end{aligned}$$

The opening round alone makes the state-versus-history gap vivid: the round automaton of Section 4 has 119,036 lattice-collapsed resolutions, but 1,888,256 order-sensitive contest records — a factor of 15.9 from flip order and deck choice alone, before bid chains and cross-round composition multiply it to 10^{34}.

10.2 Infosets, exactly

Definition 10.2 (Infoset of G)**.** An infoset of seat i is a distinct pair (public history at one of i's action points, i's private trajectory), where the trajectory is i's initial color multiset, every ordering string i has chosen at a reorder, and every discard color i has chosen. This is exactly

Table 6: The real game G, exactly. Public histories are counted by the recurrence of Section 10; per-seat infosets by Theorem 10.3. All values are exact integers computed by `real_game_size.py`; scientific notation is shown for width. The public classes row repeats Corollary 5.7's closed form $2K(H)^2 + 1$ for comparison.

	$H = 1$	$H = 2$	$H = 3$	$H = 4$	$H = 5$
Terminal public histories	32	1.217×10^6	1.522×10^{13}	4.508×10^{22}	3.023×10^{34}
Public decision histories	55	2.148×10^6	2.687×10^{13}	7.958×10^{22}	5.337×10^{34}
Boundary histories	32	1.254×10^6	1.569×10^{13}	4.647×10^{22}	3.116×10^{34}
Infosets, per seat (starter)	42	945,053	5.932×10^{12}	6.724×10^{21}	1.224×10^{33}
Infosets, both seats	82	1.890×10^6	1.186×10^{13}	1.345×10^{22}	2.448×10^{33}
Public classes ($2K^2$+1)	19	163	723	2,313	6,051

what perfect recall retains: i's own actions and observations, and nothing of the opponent's hidden ones.

Counting pairs is harder than counting histories, because a pair must be *realizable*: consistent with i's own reveals, and achievable by at least one opponent trajectory. The first constraint is a binomial prefix-consistency count (an ordering whose revealed prefix is fixed leaves $\binom{n-f}{r-p_r}$ free completions). The second is where this paper's main theorem earns its keep a second time:

Theorem 10.3 (Exact infoset counts)**.** *For heads-up Red/Black at hand size H, the number of perfect-recall infosets per seat is exactly as reported in Table 6; at $H = 5$ the opening seat has* 1,224,175,569,659,585,538,636,163,212,116,668 *infosets and the responding seat* 32 *fewer.*

Proof sketch. Walk rounds with state (own exact hand, opponent hand size, the public certainty bounds on the opponent, starter). Own-trajectory multiplicity per round is the product of the ordering-consistency binomial and the discard branching; opponent-side realizability of a candidate reveal (p_r, p_b) reduces, by Theorem 11.1 (support completeness), to non-emptiness of the folded bound interval: $\max(r_{\text{lb}}, p_r) + \max(b_{\text{lb}}, p_b) \leq n_{\text{opp}}$. Because the bounds are *exactly* the feasible set — tight as well as sound — this test admits every realizable pair and no unrealizable one, so the recurrence counts infosets exactly rather than bounding them. The recurrence is validated against explicit joint enumeration of the full game (both hands, both orderings, every deck choice) at $H = 1$ (42 and 40, exact) and $H = 2$ (945,053 and 945,049, exact). □

Remark 10.4 (The starter's edge, in counts). The two seats' infoset counts differ by exactly 2 at $H = 1$, 4 at $H = 2$, and 32 at $H = 5$: the opener acts at one node the responder never sees (the opening bid), and the difference compounds only through the thin set of lines where that node is the last asymmetric thing that happens. The game value asymmetry of Paper II has a combinatorial shadow this small.

10.3 What the numbers license

Three ratios summarize the section. The class abstraction compresses boundary histories by $\sim 10^{31}$ (3.1×10^{34} boundary histories to 5,485 reachable classes, through the intermediate 27,990 concrete boundary states of Table 8). The infoset space of G exceeds the class space by $\sim 10^{29}$. And the infoset space exceeds anything enumerable by any machine by more than ten orders of magnitude. So there will be no solve of G by tabulation, at this hand size or anywhere near it:

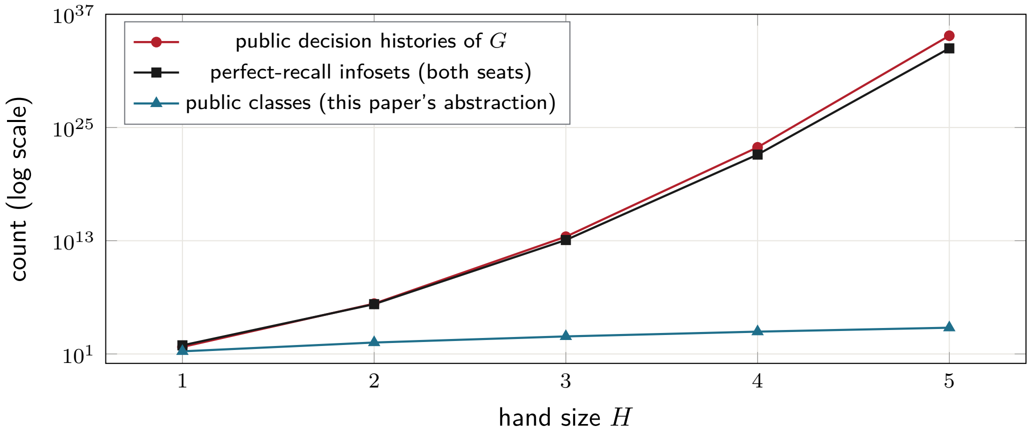

Figure 8: Thirty orders of magnitude between the game and its class abstraction. The history tree and the infoset space of G grow super-exponentially in hand size; the public class space grows like the polynomial $2\left(\binom{H+3}{3}-1\right)^2+1$. Everything the class does not carry must be reconstructed as *belief* — the subject of Section 11.

any method that engages G itself must represent histories through a summary that is small *and* strategically sufficient. The class is the right summary of public **knowledge** (Theorem 11.1); what remains — and what Section 11 takes up — is whether anything tractable is a sufficient summary of public **belief**.

11 What the Public Class Forgets

The public class of Definition 5.1 is a summary, and summaries discard things. This section says exactly what it discards, because the omissions are not incidental — they are the reason Paper II's decomposition needs care, and the raw material of Paper III.

11.1 Knowledge versus belief

There is a sharp answer to what the class keeps.

Theorem 11.1 (Support completeness)**.** *Fix a reachable class (ℓ, s_0, s_1). The hand compositions consistent with some history reaching it are exactly*

$$\left\{(r_i, b_i)\ :\ r_i \geq r^i_{\mathrm{lb}},\ \ b_i \geq b^i_{\mathrm{lb}},\ \ r_i + b_i = n_i,\ \ i = 0, 1\right\}.$$

That is, the certainty bounds are simultaneously sound (nothing outside the box is possible) and tight (everything inside it occurs).

Soundness is Lemma 5.2. Tightness is verified exhaustively: across all 5,485 reachable classes at $H = 5$, the hand compositions actually realized by the closure of Section 9.2 fill the box exactly, with no class showing a gap.

So the class is a *complete* record of what is publicly **known**. It is not a record of what is publicly **believed**. Knowing the support of a distribution is not knowing the distribution, and the gap between the two is large enough to matter.

11.2 Two paths to the same class

Consider a seat that ends up holding four cards, certainly including at least one red — public statistic $(4, 1, 0)$. Two histories reach it:

Path A (decay). The seat held five cards; *two reds were flipped off its deck* in round 1. It lost, and discarded. The bound rose to 2, then decayed by one: $r_{\text{lb}} = 1$, $n = 4$.

Path B (fresh sighting). The seat held five cards and nothing was revealed from it in round 1. It lost and discarded, leaving four cards and no bounds. In round 2 *one red* was flipped off its deck: $r_{\text{lb}} = 1$, $n = 4$.

Identical classes. But an observer who watched Path A saw two reds go into that hand and only one card leave it, so it very likely still holds two. An observer of Path B saw one red and nothing else. Table 7 quantifies the gap under a neutral reference policy — uniformly random ordering and uniformly random discard — chosen only to put numbers on it.

Table 7: Posterior over the reds a seat holds, given the *same* public statistic $(n, r_{\text{lb}}, b_{\text{lb}}) = (4, 1, 0)$. Both paths certify "at least one red" and neither certifies more — the bound is tight for each — yet the distributions differ substantially.

Reds held	Path A (two seen, then discard)	Path B (discard, then one seen)
1	0.0531	0.1248
2	0.3184	0.3902
3	0.4521	0.3745
4	0.1765	0.1104
Expected reds	2.752	2.471
Pr[at least 2 reds]	0.947	0.875

The mechanism is the max-fold of Lemma 5.2. Folding is *lossy in a specific direction*: it remembers the height of the bound but forgets whether that height is fresh evidence or the residue of a taller bound worn down by discards. A bound of 1 that used to be 3 is a much stronger signal than a bound of 1 that has always been 1, and the class cannot tell them apart.

11.3 The bid history

The second omission is blunter: a round's bid history is cleared when the round ends (Section 2.3) and appears nowhere in the class. Yet bids are chosen by players who can see their own cards, so a bid history is evidence about the bidder's hand, and a *sequence* of bid histories is evidence about how that player bids — how readily they bluff, how they price a raise.

Neither kind of evidence survives into the public class. Note the asymmetry with the flip record: flips are physical reveals and enter the class as bounds; bids are cheap talk and enter not at all. A player who opened 5R last round and was caught has told the table something real, and the class forgets it completely.

11.4 The reachable belief family

The two omissions above say what beliefs the class *loses*. This subsection proves the surprising positive counterpart: the beliefs themselves — over both seats' private situations jointly, under *any* strategy profile — live in a family whose shape is a property of the game, not of the profile.

Write seat i's private *type* at a boundary as $t_i = (r_i, x_i)$: the reds currently held and the reds among i's own discards. (This is the private statistic the companion papers solve over; the pending material here is only its structure.)

Lemma 11.2 (Deal invariance)**.** *Let $a_i = r_i + x_i$. Then a_i is constant along every play: it equals the number of reds dealt to seat i. Flips do not change it (revealed cards stay in the hand), a red discard moves one unit from r_i to x_i, and a black discard changes neither.*

Theorem 11.3 (Rank-one factorization of reachable beliefs)**.** *Fix any strategy profile and any public history h ending at a boundary of class κ. The joint posterior over type pairs factorizes as*

$$\mu_h(t_0, t_1) \;\propto\; \lambda_\kappa(t_0, t_1) \cdot g_0(t_0) \cdot g_1(t_1),$$

where λ_κ is the deal-and-bounds prior of class κ and g_i is a per-seat nonnegative weight (a likelihood tilt) depending only on seat i's strategy and h.

Proof. All correlation between the seats' private situations originates in the deal, and by Lemma 11.2 the deal enters the type pair only through (a_0, a_1), which is a deterministic function of the types themselves — so the deal coupling is absorbed into λ_κ and rides along every transition unchanged. Conditional on types, each seat's observable actions depend only on its own type and h, so the likelihood of h is a product of per-seat terms. Bayes' rule then yields the displayed form. □

Corollary 11.4 (Model-free conditionals)**.** *The conditional law of seat i's type given the opponent's type, $\mu_h(t_i \mid t_{1-i})$, does not depend on g_{1-i}: any positive weighting of opponent types produces the same conditionals. An observer can be entirely wrong about how the opponent plays and still hold exactly correct own-side conditionals.*

Theorem 11.3 was found empirically before it was proved: over 17,140 boundaries of 3,000 self-play games under the solved profile of the companion papers, the measured rank-one residual of the exact posterior is 3.3×10^{-16} — floating-point zero — with no exceptions. The theorem is load-bearing there twice over: it makes exact belief tracking cheap (two per-seat vectors instead of a joint matrix), and Corollary 11.4 is what lets a re-solving agent keep exact own-side conditionals even against an unmodeled opponent (Paper III's filter-repair construction).

Together with Section 10, this locates the game's true sufficient statistic. The public class is the strategy-independent *knowledge*; the reachable *belief* state is the class plus two per-seat tilt vectors (g_0, g_1) — a finite-dimensional family, against the 10^{34} raw histories that induce it. Strategy dependence enters only through where in the family a profile lands, never through the family's shape.

11.5 Measured: how much the class forgets

The gap of Table 7 was a two-path worked example. Under the solved profile of the companion papers it has been measured wholesale: the reach-weighted total-variation distance between the class prior λ_κ and the true public posterior μ_h is **0.842** over self-play boundaries, rising monotonically with depth from 0.66 (first boundary) to 0.95 (endgame layers). Replacing the prior with the best possible *class-keyed* statistic closes only about 17 points of that gap, and small public-statistic enrichments (e.g. freshness bits distinguishing Path A from Path B above) add only 2–3 more: most of the belief lives in within-class variation that no small public statistic reaches. These are profile-dependent numbers — one representative profile, the solved one — but they put a scale on the qualitative claim of this section: the class prior at depth is nearly orthogonal to the belief a competent observer actually holds. The companion papers carry the consequences.

11.6 Why the class stops where it does

It would be natural to ask for a finer public state, and worth seeing why one is not simply available.

The exact public posterior over hands depends on the players' *strategies*. If I know you always discard a black when you hold one, watching you discard is enormously informative; if I know nothing about your policy, the same observation tells me almost nothing. So the exact posterior is not a function of the public history alone — it is a function of the history *and* a model of the opponent. It cannot be tabulated as part of a description of the game, because it is not a property of the game.

The certainty bounds are what remains once that dependence is stripped out. They are precisely the facts entailed by the rules and the public record regardless of how anyone has been playing, which is why they, and not some richer statistic, are the right stopping point for a paper about the game itself. Theorem 11.1 is the formal version of that claim: the class is exactly the strategy-independent content of the public history.

Everything beyond it — the fold's path dependence, the bid history, the opponent's discard tendencies — is belief rather than knowledge. What this paper can still say about belief without committing to a strategy model is Section 11.4: the reachable belief family has a game-determined shape, $\lambda_\kappa \cdot g_0 \cdot g_1$, and only the tilts (g_0, g_1) are anyone's model of anyone. The class is the complete knowledge state; the class plus the tilts is the belief state; and the companion papers are, respectively, what happens when a solver uses only the first (Paper II) and what it takes to responsibly use the second (Paper III).

12 Summary of Counts

Table 8 gathers every quantity derived above across the three games. Reading it left to right shows how sharply the game grows with hand size: the deal count climbs by nine orders of magnitude, and the opening round's contest structure by four, while the *public* state space — the thing an observer must actually track — grows only from 19 to 6,051. That gap is the point of Section 5: the public projection is exponentially smaller than the machine it summarizes, and it is what the companion paper's algorithms index on.

13 What Comes Next

This paper stops at the dynamics and the information structure. Two companion papers carry the threads forward, and it is worth being precise about which thread each one takes.

Paper II solves the class abstraction, exactly, and measures the cost. Section 10 rules out engaging the history tree directly, and Theorem 11.1 makes the class the canonical small summary — so Paper II decomposes the game at boundaries, re-derives entry beliefs from the

Table 8: Master table. Every entry is produced by `state_counts.py` and, where an engine counterpart exists, verified against the shipped implementation by `verify_against_engine.py`.

Quantity	$H = 1$	$H = 2$	$H = 5$
Card-level deals	2,652	1,624,350	3,986,646,103,440
Color-level deal outcomes	4	9	36
Rounds (min–max)	1–1	2–3	5–9
Opening-round bid states	7	31	2,047
Opening-round procurement	14	190	65,790
Opening-round resolutions	28	348	119,036
Per-seat public statistic $K(H)$	3	9	55
Public classes $2K^2 + 1$	19	163	6,051
of which terminal	18	54	330
of which reachable	17	149	5,485
Concrete boundary states	28	345	27,990
Public decision histories of G	55	2,147,639	5.337×10^{34}
Perfect-recall infosets of G (both seats)	82	1,890,102	2.448×10^{33}

class prior λ_κ, and solves every one of the 5,721 subgame-bearing classes with CFR+. That belief re-derivation is exactly the approximation Sections 11.4 and 11.5 put a scale on, and Paper II's second half measures what it costs in play: the abstraction's equilibrium is exploitable in the real game in ways its own convergence meters cannot see, including a converged action provably dominated in G.

Paper III uses the belief family this paper identified. Theorem 11.3 says the reachable beliefs of G form the finite family $\lambda_\kappa \cdot g_0 \cdot g_1$; Paper III tracks the tilts exactly with a public filter, re-solves each boundary under the true posterior instead of the prior, and pursues a *certified upper bound* on real-game exploitability via safe re-solving. Corollary 11.4 is load-bearing there: own-side conditionals survive an arbitrarily wrong opponent model.

Three technical threads from this paper recur in both.

From public classes to information sets. A player at a boundary knows their public class *and* their own hand. The private component turns out to need two numbers, not one: the reds currently held, and the reds among the cards that player has itself discarded. The second is private information about the *deck*, not about the hand — which brings us to the next thread.

The finite deck is load-bearing. Because the deck is finite and balanced, your own cards tell you about your opponent's. Seeing k reds among your own five shifts the chance an unseen card is red from 1/2 to $(26 - k)/47$, a swing from 0.447 to 0.553. In the one-card game this is the *entire* strategic content: the opener's only real decision is whether to bet that the two cards match or differ, and it should use its own five dealt cards to choose. That game's exact value is $\frac{40651}{78302} \approx 0.5192$ to the opener — above 1/2 purely because of sampling without replacement, and above the $26/51 \approx 0.5098$ that an opener ignoring its own cards would get. Under an idealised infinite deck the same game is exactly even.

Composition is the hard part. The layered DAG of Section 9 invites solving the game layer by layer, bootstrapping each class from the ones below. Two hazards, both visible from this paper, make that harder than it looks. First, by Section 11 the class is not a sufficient statistic for beliefs, so re-deriving a belief from the class at a layer boundary is an approximation, not an identity — it silently merges the two paths of Table 7. Second, whether the decomposition is faithful depends on where the layers are cut relative to the private discard decision. Getting either wrong is not a small error. Both are the subject of Paper II — the second cost an early solver generation a measured mispricing, and the first is the origin of everything Paper III exists to repair.

A Reproducing the numbers

Two scripts accompany this paper.

`state_counts.py` re-derives every class-space count from the rules, independently of the game engine, and prints the corresponding tables. It includes the forward-closure search of Section 9.

`real_game_size.py` computes every count in Section 10 — histories, decision histories, boundary histories, and the exact per-seat infoset counts — as exact big integers, and runs the brute-force joint-enumeration validation of Theorem 10.3 at $H = 1, 2$. Pure standard library.

`verify_against_engine.py` cross-checks the round-automaton counts, the class-space counts and the reachable set against the shipped implementation, so any disagreement is a genuine bug in one of the two rather than a difference of convention.

Run them with

```
cd research && python3 state_counts.py
cd research && python3 real_game_size.py
cd research && ../simulator/.venv/bin/python verify_against_engine.py
```

B Glossary

Boundary The moment a contest resolves: loser known, discard not yet made. The natural cut point for describing the public state.

Chain The strictly increasing sequence of counts bid in a round.

Class The public state at a boundary (Definition 5.1).

Layer T Total cards in play after the pending discard; strictly decreasing, hence a valid induction index.

Procurement The contest phase: the bidder flips cards trying to produce the claim.

Sighting The multiset of colors revealed from one seat's deck in one round.